Writing Exercises

110. Frida started to solve the equation $-3x = 36$ by adding 3 to both sides. Explain why Frida's method will result in the correct solution.

111. Emiliano thinks $x = 40$ is the solution to the equation $\frac{1}{2}x = 80$. Explain why he is wrong.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
solve equations using the Division and Multiplication Properties of Equality.			
solve equations that need to be simplified.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

8.3 Solve Equations with Variables and Constants on Both Sides

Learning Objectives

By the end of this section, you will be able to:

- Solve an equation with constants on both sides
- Solve an equation with variables on both sides
- Solve an equation with variables and constants on both sides
- Solve equations using a general strategy

☑ **BE PREPARED** 8.7 Before you get started, take this readiness quiz.

Simplify: $4y - 9 + 9$.
If you missed this problem, review Example 2.22.

☑ **BE PREPARED** 8.8 Solve: $y + 12 = 16$.
If you missed this problem, review Example 2.31.

☑ **BE PREPARED** 8.9 Solve: $-3y = 63$.
If you missed this problem, review Example 3.65.

Solve an Equation with Constants on Both Sides

You may have noticed that in all the equations we have solved so far, all the variable terms were on only one side of the equation with the constants on the other side. This does not happen all the time—so now we'll see how to solve equations where the variable terms and/or constant terms are on both sides of the equation.

Our strategy will involve choosing one side of the equation to be the variable side, and the other side of the equation to be the constant side. Then, we will use the Subtraction and Addition Properties of Equality, step by step, to get all the variable terms together on one side of the equation and the constant terms together on the other side.

By doing this, we will transform the equation that started with variables and constants on both sides into the form $ax = b$. We already know how to solve equations of this form by using the Division or Multiplication Properties of Equality.

EXAMPLE 8.20

Solve: $4x + 6 = -14$.

Solution

In this equation, the variable is only on the left side. It makes sense to call the left side the variable side. Therefore, the right side will be the constant side. We'll write the labels above the equation to help us remember what goes where.

	variable constant $4x + 6 = -14$
Since the left side is the variable side, the 6 is out of place. We must "undo" adding 6 by subtracting 6, and to keep the equality we must subtract 6 from both sides. Use the Subtraction Property of Equality.	$4x + 6 - 6 = -14 - 6$
Simplify.	$4x = -20$
Now all the xs are on the left and the constant on the right.	
Use the Division Property of Equality.	$\frac{4x}{4} = \frac{-20}{4}$
Simplify.	$x = -5$
Check:	$4x + 6 = -14$
Let $x = -5$.	$4(-5) + 6 = -14$
	$-20 + 6 = -14$
	$-14 = -14$ ✓

TRY IT 8.39 Solve: $3x + 4 = -8$.

TRY IT 8.40 Solve: $5a + 3 = -37$.

EXAMPLE 8.21

Solve: $2y - 7 = 15$.

Solution

Notice that the variable is only on the left side of the equation, so this will be the variable side and the right side will be the constant side. Since the left side is the variable side, the 7 is out of place. It is subtracted from the $2y$, so to 'undo' subtraction, add 7 to both sides.

	variable constant $2y - 7 = 15$
Add 7 to both sides.	$2y - 7 + 7 = 15 + 7$
Simplify.	$2y = 22$
The variables are now on one side and the constants on the other.	
Divide both sides by 2.	$\frac{2y}{2} = \frac{22}{2}$

Simplify.		$y = 11$
Check:	$2y - 7 = 15$	
Substitute: $y = 11$.	$2 \cdot 11 - 7 \stackrel{?}{=} 15$	
	$22 - 7 \stackrel{?}{=} 15$	
	$15 = 15$ ✓	

TRY IT 8.41 Solve: $5y - 9 = 16$.

TRY IT 8.42 Solve: $3m - 8 = 19$.

Solve an Equation with Variables on Both Sides

What if there are variables on both sides of the equation? We will start like we did above—choosing a variable side and a constant side, and then use the Subtraction and Addition Properties of Equality to collect all variables on one side and all constants on the other side. Remember, what you do to the left side of the equation, you must do to the right side too.

EXAMPLE 8.22

Solve: $5x = 4x + 7$.

Solution

Here the variable, x, is on both sides, but the constants appear only on the right side, so let's make the right side the "constant" side. Then the left side will be the "variable" side.

		variable constant $5x = 4x + 7$
We don't want any variables on the right, so subtract the $4x$.		$5x - 4x = 4x - 4x + 7$
Simplify.		$x = 7$
We have all the variables on one side and the constants on the other. We have solved the equation.		
Check:	$5x = 4x + 7$	
Substitute 7 for x.	$5(7) \stackrel{?}{=} 4(7) + 7$	
	$35 \stackrel{?}{=} 28 + 7$	
	$35 = 35$ ✓	

TRY IT 8.43 Solve: $6n = 5n + 10$.

TRY IT 8.44 Solve: $-6c = -7c + 1$.

EXAMPLE 8.23

Solve: $5y - 8 = 7y$.

Solution

The only constant, -8, is on the left side of the equation and variable, y, is on both sides. Let's leave the constant on the left and collect the variables to the right.

	$\overset{\text{constant}}{5y - 8} = \overset{\text{variable}}{7y}$
Subtract $5y$ from both sides.	$5y - 5y - 8 = 7y - 5y$
Simplify.	$-8 = 2y$
We have the variables on the right and the constants on the left. Divide both sides by 2.	$\frac{-8}{2} = \frac{2y}{2}$
Simplify.	$-4 = y$
Rewrite with the variable on the left.	$y = -4$
Check: Let $y = -4$.	
$5y - 8 = 7y$	
$5(-4) - 8 \stackrel{?}{=} 7(-4)$	
$-20 - 8 \stackrel{?}{=} -28$	
$-28 = -28$ ✓	

TRY IT 8.45 Solve: $3p - 14 = 5p$.

TRY IT 8.46 Solve: $8m + 9 = 5m$.

EXAMPLE 8.24

Solve: $7x = -x + 24$.

Solution

The only constant, 24, is on the right, so let the left side be the variable side.

	$\overset{\text{variable side}}{7x} = \overset{\text{constant side}}{-x + 24}$
Remove the $-x$ from the right side by adding x to both sides.	$7x + x = -x + x + 24$

Simplify.	$8x = 24$
All the variables are on the left and the constants are on the right. Divide both sides by 8.	$\frac{8x}{8} = \frac{24}{8}$
Simplify.	$x = 3$
Check: Substitute $x = 3$.	
$7x = -x + 24$ $7(3) \stackrel{?}{=} -(3) + 24$ $21 = 21$ ✓	

TRY IT 8.47 Solve: $12j = -4j + 32$.

TRY IT 8.48 Solve: $8h = -4h + 12$.

Solve Equations with Variables and Constants on Both Sides

The next example will be the first to have variables *and* constants on both sides of the equation. As we did before, we'll collect the variable terms to one side and the constants to the other side.

EXAMPLE 8.25

Solve: $7x + 5 = 6x + 2$.

Solution

Start by choosing which side will be the variable side and which side will be the constant side. The variable terms are $7x$ and $6x$. Since 7 is greater than 6, make the left side the variable side and so the right side will be the constant side.

	$7x + 5 = 6x + 2$
Collect the variable terms to the left side by subtracting $6x$ from both sides.	$7x - 6x + 5 = 6x - 6x + 2$
Simplify.	$x + 5 = 2$
Now, collect the constants to the right side by subtracting 5 from both sides.	$x + 5 - 5 = 2 - 5$
Simplify.	$x = -3$
The solution is $x = -3$.	
Check: Let $x = -3$.	
$7x + 5 = 6x + 2$ $7(-3) + 5 \stackrel{?}{=} 6(-3) + 2$ $-21 + 5 \stackrel{?}{=} -18 + 2$ $-16 = -16$ ✓	

TRY IT 8.49 Solve: $12x + 8 = 6x + 2$.

TRY IT 8.50 Solve: $9y + 4 = 7y + 12$.

We'll summarize the steps we took so you can easily refer to them.

HOW TO

Solve an equation with variables and constants on both sides.

Step 1. Choose one side to be the variable side and then the other will be the constant side.
Step 2. Collect the variable terms to the variable side, using the Addition or Subtraction Property of Equality.
Step 3. Collect the constants to the other side, using the Addition or Subtraction Property of Equality.
Step 4. Make the coefficient of the variable 1, using the Multiplication or Division Property of Equality.
Step 5. Check the solution by substituting it into the original equation.

It is a good idea to make the variable side the one in which the variable has the larger coefficient. This usually makes the arithmetic easier.

EXAMPLE 8.26

Solve: $6n - 2 = -3n + 7$.

Solution

We have $6n$ on the left and $-3n$ on the right. Since $6 > -3$, make the left side the "variable" side.

	$6n - 2 = -3n + 7$
We don't want variables on the right side—add $3n$ to both sides to leave only constants on the right.	$6n + 3n - 2 = -3n + 3n + 7$
Combine like terms.	$9n - 2 = 7$
We don't want any constants on the left side, so add 2 to both sides.	$9n - 2 + 2 = 7 + 2$
Simplify.	$9n = 9$
The variable term is on the left and the constant term is on the right. To get the coefficient of n to be one, divide both sides by 9.	$\frac{9n}{9} = \frac{9}{9}$
Simplify.	$n = 1$
Check: Substitute 1 for n.	
$6n - 2 = -3n + 7$ $6(1) - 2 \stackrel{?}{=} -3(1) + 7$ $4 = 4$ ✓	

TRY IT 8.51 Solve: $8q - 5 = -4q + 7$.

TRY IT 8.52 Solve: $7n - 3 = n + 3$.

EXAMPLE 8.27

Solve: $2a - 7 = 5a + 8$.

Solution

This equation has $2a$ on the left and $5a$ on the right. Since $5 > 2$, make the right side the variable side and the left side the constant side.

	$2a - 7 = 5a + 8$
Subtract $2a$ from both sides to remove the variable term from the left.	$2a - 2a - 7 = 5a - 2a + 8$
Combine like terms.	$-7 = 3a + 8$
Subtract 8 from both sides to remove the constant from the right.	$-7 - 8 = 3a + 8 - 8$
Simplify.	$-15 = 3a$
Divide both sides by 3 to make 1 the coefficient of a.	$\frac{-15}{3} = \frac{3a}{3}$
Simplify.	$-5 = a$
Check: Let $a = -5$.	
$2a - 7 = 5a + 8$ $2(-5) - 7 \stackrel{?}{=} 5(-5) + 8$ $-10 - 7 \stackrel{?}{=} -25 + 8$ $-17 = -17$ ✓	

Note that we could have made the left side the variable side instead of the right side, but it would have led to a negative coefficient on the variable term. While we could work with the negative, there is less chance of error when working with positives. The strategy outlined above helps avoid the negatives!

TRY IT 8.53 Solve: $2a - 2 = 6a + 18$.

TRY IT 8.54 Solve: $4k - 1 = 7k + 17$.

To solve an equation with fractions, we still follow the same steps to get the solution.

EXAMPLE 8.28

Solve: $\frac{3}{2}x + 5 = \frac{1}{2}x - 3$.

Solution

Since $\frac{3}{2} > \frac{1}{2}$, make the left side the variable side and the right side the constant side.

	$\frac{3}{2}x+5=\frac{1}{2}x-3$
Subtract $\frac{1}{2}x$ from both sides.	$\frac{3}{2}x-\frac{1}{2}x+5=\frac{1}{2}x-\frac{1}{2}x-3$
Combine like terms.	$x+5=-3$
Subtract 5 from both sides.	$x+5-5=-3-5$
Simplify.	$x=-8$
Check: Let $x=-8$.	
$\frac{3}{2}x+5=\frac{1}{2}x-3$ $\frac{3}{2}(-8)+5\stackrel{?}{=}\frac{1}{2}(-8)-3$ $-12+5\stackrel{?}{=}-4-3$ $-7=-7$ ✓	

TRY IT 8.55 Solve: $\frac{7}{8}x-12=-\frac{1}{8}x-2$.

TRY IT 8.56 Solve: $\frac{7}{6}y+11=\frac{1}{6}y+8$.

We follow the same steps when the equation has decimals, too.

EXAMPLE 8.29

Solve: $3.4x+4=1.6x-5$.

Solution

Since $3.4>1.6$, make the left side the variable side and the right side the constant side.

	$3.4x+4=1.6x-5$
Subtract $1.6x$ from both sides.	$3.4x-1.6x+4=1.6x-1.6x-5$
Combine like terms.	$1.8x+4=-5$
Subtract 4 from both sides.	$1.8x+4-4=-5-4$
Simplify.	$1.8x=-9$
Use the Division Property of Equality.	$\frac{1.8x}{1.8}=\frac{-9}{1.8}$
Simplify.	$x=-5$

Check: Let $x = -5$.

$$3.4x + 4 = 1.6x - 5$$
$$3.4(-5) + 4 \stackrel{?}{=} 1.6(-5) - 5$$
$$-17 + 4 \stackrel{?}{=} -8 - 5$$
$$-13 = -13 \checkmark$$

TRY IT 8.57 Solve: $2.8x + 12 = -1.4x - 9$.

TRY IT 8.58 Solve: $3.6y + 8 = 1.2y - 4$.

Solve Equations Using a General Strategy

Each of the first few sections of this chapter has dealt with solving one specific form of a linear equation. It's time now to lay out an overall strategy that can be used to solve *any* linear equation. We call this the *general strategy*. Some equations won't require all the steps to solve, but many will. Simplifying each side of the equation as much as possible first makes the rest of the steps easier.

HOW TO

Use a general strategy for solving linear equations.

Step 1. Simplify each side of the equation as much as possible. Use the Distributive Property to remove any parentheses. Combine like terms.
Step 2. Collect all the variable terms to one side of the equation. Use the Addition or Subtraction Property of Equality.
Step 3. Collect all the constant terms to the other side of the equation. Use the Addition or Subtraction Property of Equality.
Step 4. Make the coefficient of the variable term to equal to 1. Use the Multiplication or Division Property of Equality. State the solution to the equation.
Step 5. Check the solution. Substitute the solution into the original equation to make sure the result is a true statement.

EXAMPLE 8.30

Solve: $3(x + 2) = 18$.

Solution

	$3(x + 2) = 18$
Simplify each side of the equation as much as possible. Use the Distributive Property.	$3x + 6 = 18$
Collect all variable terms on one side of the equation—all xs are already on the left side.	
Collect constant terms on the other side of the equation. Subtract 6 from each side	$3x + 6 - 6 = 18 - 6$
Simplify.	$3x = 12$

Make the coefficient of the variable term equal to 1. Divide each side by 3.	$\frac{3x}{3} = \frac{12}{3}$
Simplify.	$x = 4$
Check: Let $x = 4$.	
$3(x+2) = 18$ $3(4+2) \stackrel{?}{=} 18$ $3(6) \stackrel{?}{=} 18$ $18 = 18$ ✓	

TRY IT 8.59 Solve: $5\,(x+3) = 35$.

TRY IT 8.60 Solve: $6\,(y-4) = -18$.

EXAMPLE 8.31

Solve: $-(x+5) = 7$.

Solution

	$-(x+5) = 7$
Simplify each side of the equation as much as possible by distributing. The only x term is on the left side, so all variable terms are on the left side of the equation.	$-x - 5 = 7$
Add 5 to both sides to get all constant terms on the right side of the equation.	$-x - 5 + 5 = 7 + 5$
Simplify.	$-x = 12$
Make the coefficient of the variable term equal to 1 by multiplying both sides by -1.	$-1(-x) = -1(12)$
Simplify.	$x = -12$
Check: Let $x = -12$.	
$-(x+5) = 7$ $-(-12+5) \stackrel{?}{=} 7$ $-(-7) \stackrel{?}{=} 7$ $7 = 7$ ✓	

TRY IT 8.61 Solve: $-(y+8) = -2$.

TRY IT 8.62 Solve: $-(z+4) = -12$.

EXAMPLE 8.32

Solve: $4\,(x-2)+5=-3$.

Solution

	$4(x-2)+5=-3$
Simplify each side of the equation as much as possible. Distribute.	$4x-8+5=-3$
Combine like terms	$4x-3=-3$
The only x is on the left side, so all variable terms are on one side of the equation.	
Add 3 to both sides to get all constant terms on the other side of the equation.	$4x-3+3=-3+3$
Simplify.	$4x=0$
Make the coefficient of the variable term equal to 1 by dividing both sides by 4.	$\frac{4x}{4}=\frac{0}{4}$
Simplify.	$x=0$
Check: Let $x=0$.	
$4(x-2)+5=-3$ $4(0-2)+5\stackrel{?}{=}-3$ $4(-2)+5\stackrel{?}{=}-3$ $-8+5\stackrel{?}{=}-3$ $-3=-3$ ✓	

TRY IT 8.63 Solve: $2\,(a-4)+3=-1$.

TRY IT 8.64 Solve: $7\,(n-3)-8=-15$.

EXAMPLE 8.33

Solve: $8-2\,(3y+5)=0$.

Solution

Be careful when distributing the negative.

	$8-2(3y+5)=0$
Simplify—use the Distributive Property.	$8-6y-10=0$
Combine like terms.	$-6y-2=0$
Add 2 to both sides to collect constants on the right.	$-6y-2+2=0+2$

Simplify.	$-6y = 2$
Divide both sides by –6.	$\frac{-6y}{-6} = \frac{2}{-6}$
Simplify.	$y = -\frac{1}{3}$
Check: Let $y = -\frac{1}{3}$.	
$8 - 2(3y + 5) = 0$ $8 - 2\left[3\left(-\frac{1}{3}\right) + 5\right] = 0$ $8 - 2(-1 + 5) \stackrel{?}{=} 0$ $8 - 2(4) \stackrel{?}{=} 0$ $8 - 8 \stackrel{?}{=} 0$ $0 = 0$ ✓	

TRY IT 8.65 Solve: $12 - 3(4j + 3) = -17$.

TRY IT 8.66 Solve: $-6 - 8(k - 2) = -10$.

EXAMPLE 8.34

Solve: $3(x - 2) - 5 = 4(2x + 1) + 5$.

Solution

	$3(x - 2) - 5 = 4(2x + 1) + 5$
Distribute.	$3x - 6 - 5 = 8x + 4 + 5$
Combine like terms.	$3x - 11 = 8x + 9$
Subtract $3x$ to get all the variables on the right since $8 > 3$.	$3x - 3x - 11 = 8x - 3x + 9$
Simplify.	$-11 = 5x + 9$
Subtract 9 to get the constants on the left.	$-11 - 9 = 5x + 9 - 9$
Simplify.	$-20 = 5x$
Divide by 5.	$\frac{-20}{5} = \frac{5x}{5}$
Simplify.	$-4 = x$

Check: Substitute: $-4 = x$.

$$
\begin{aligned}
3(x-2)-5 &= 4(2x+1)+5\\
3(-4-2)-5 &\stackrel{?}{=} 4[2(-4)+1]+5\\
3(-6)-5 &\stackrel{?}{=} 4(-8+1)+5\\
-18-5 &\stackrel{?}{=} 4(-7)+5\\
-23 &\stackrel{?}{=} -28+5\\
-23 &= -23\ \checkmark
\end{aligned}
$$

TRY IT 8.67 Solve: $6(p-3)-7 = 5(4p+3)-12$.

TRY IT 8.68 Solve: $8(q+1)-5 = 3(2q-4)-1$.

EXAMPLE 8.35

Solve: $\frac{1}{2}(6x-2) = 5-x$.

Solution

	$\frac{1}{2}(6x-2) = 5-x$
Distribute.	$3x-1 = 5-x$
Add x to get all the variables on the left.	$3x-1+x = 5-x+x$
Simplify.	$4x-1 = 5$
Add 1 to get constants on the right.	$4x-1+1 = 5+1$
Simplify.	$4x = 6$
Divide by 4.	$\frac{4x}{4} = \frac{6}{4}$
Simplify.	$x = \frac{3}{2}$

Check: Let $x = \frac{3}{2}$.

$$\frac{1}{2}(6x-2) = 5 - x$$
$$\frac{1}{2}\left(6 \cdot \frac{3}{2} - 2\right) \stackrel{?}{=} 5 - \frac{3}{2}$$
$$\frac{1}{2}(9-2) \stackrel{?}{=} \frac{10}{2} - \frac{3}{2}$$
$$\frac{1}{2}(7) \stackrel{?}{=} \frac{7}{2}$$
$$\frac{7}{2} = \frac{7}{2} \checkmark$$

TRY IT 8.69 Solve: $\frac{1}{3}(6u+3) = 7 - u$.

TRY IT 8.70 Solve: $\frac{2}{3}(9x-12) = 8 + 2x$.

In many applications, we will have to solve equations with decimals. The same general strategy will work for these equations.

EXAMPLE 8.36

Solve: $0.24(100x+5) = 0.4(30x+15)$.

Solution

	$0.24(100x+5) = 0.4(30x+15)$
Distribute.	$24x + 1.2 = 12x + 6$
Subtract $12x$ to get all the xs to the left.	$24x + 1.2 - 12x = 12x + 6 - 12x$
Simplify.	$12x + 1.2 = 6$
Subtract 1.2 to get the constants to the right.	$12x + 1.2 - 1.2 = 6 - 1.2$
Simplify.	$12x = 4.8$
Divide.	$\frac{12x}{12} = \frac{4.8}{12}$
Simplify.	$x = 0.4$

Check: Let $x = 0.4$.

$$0.24(100x + 5) = 0.4(30x + 15)$$
$$0.24(100(0.4) + 5) \stackrel{?}{=} 0.4(30(0.4) + 15)$$
$$0.24(40 + 5) \stackrel{?}{=} 0.4(12 + 15)$$
$$0.24(45) \stackrel{?}{=} 0.4(27)$$
$$10.8 = 10.8 \checkmark$$

TRY IT 8.71 Solve: $0.55(100n + 8) = 0.6(85n + 14)$.

TRY IT 8.72 Solve: $0.15(40m - 120) = 0.5(60m + 12)$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Solving Multi-Step Equations (https://www.openstax.org/l/24SolveMultStep)
Solve an Equation with Variable Terms on Both Sides (https://www.openstax.org/l/24SolveEquatVar)
Solving Multi-Step Equations (L5.4) (https://www.openstax.org/l/24MultiStepEqu)
Solve an Equation with Variables and Parentheses on Both Sides (https://www.openstax.org/l/24EquVarParen)

SECTION 8.3 EXERCISES

Practice Makes Perfect

Solve an Equation with Constants on Both Sides

In the following exercises, solve the equation for the variable.

112. $6x - 2 = 40$ **113.** $7x - 8 = 34$ **114.** $11w + 6 = 93$

115. $14y + 7 = 91$ **116.** $3a + 8 = -46$ **117.** $4m + 9 = -23$

118. $-50 = 7n - 1$ **119.** $-47 = 6b + 1$ **120.** $25 = -9y + 7$

121. $29 = -8x - 3$ **122.** $-12p - 3 = 15$ **123.** $-14\mathrm{q} - 15 = 13$

Solve an Equation with Variables on Both Sides

In the following exercises, solve the equation for the variable.

124. $8z = 7z - 7$ **125.** $9k = 8k - 11$ **126.** $4x + 36 = 10x$

127. $6x + 27 = 9x$ **128.** $c = -3c - 20$ **129.** $b = -4b - 15$

130. $5q = 44 - 6q$ **131.** $7z = 39 - 6z$ **132.** $3y + \frac{1}{2} = 2y$

133. $8x + \frac{3}{4} = 7x$ **134.** $-12a - 8 = -16a$ **135.** $-15r - 8 = -11r$

Solve an Equation with Variables and Constants on Both Sides

In the following exercises, solve the equations for the variable.

136. $6x - 15 = 5x + 3$ **137.** $4x - 17 = 3x + 2$ **138.** $26 + 8d = 9d + 11$

139. $21 + 6f = 7f + 14$ **140.** $3p - 1 = 5p - 33$ **141.** $8q - 5 = 5q - 20$

142. $4a + 5 = -a - 40$ **143.** $9c + 7 = -2c - 37$ **144.** $8y - 30 = -2y + 30$

145. $12x - 17 = -3x + 13$

146. $2z - 4 = 23 - z$

147. $3y - 4 = 12 - y$

148. $\frac{5}{4}c - 3 = \frac{1}{4}c - 16$

149. $\frac{4}{3}m - 7 = \frac{1}{3}m - 13$

150. $8 - \frac{2}{5}q = \frac{3}{5}q + 6$

151. $11 - \frac{1}{4}a = \frac{3}{4}a + 4$

152. $\frac{4}{3}n + 9 = \frac{1}{3}n - 9$

153. $\frac{5}{4}a + 15 = \frac{3}{4}a - 5$

154. $\frac{1}{4}y + 7 = \frac{3}{4}y - 3$

155. $\frac{3}{5}p + 2 = \frac{4}{5}p - 1$

156. $14n + 8.25 = 9n + 19.60$

157. $13z + 6.45 = 8z + 23.75$

158. $2.4w - 100 = 0.8w + 28$

159. $2.7w - 80 = 1.2w + 10$

160. $5.6r + 13.1 = 3.5r + 57.2$

161. $6.6x - 18.9 = 3.4x + 54.7$

Solve an Equation Using the General Strategy

In the following exercises, solve the linear equation using the general strategy.

162. $5(x + 3) = 75$

163. $4(y + 7) = 64$

164. $8 = 4(x - 3)$

165. $9 = 3(x - 3)$

166. $20(y - 8) = -60$

167. $14(y - 6) = -42$

168. $-4(2n + 1) = 16$

169. $-7(3n + 4) = 14$

170. $3(10 + 5r) = 0$

171. $8(3 + 3p) = 0$

172. $\frac{2}{3}(9c - 3) = 22$

173. $\frac{3}{5}(10x - 5) = 27$

174. $5(1.2u - 4.8) = -12$

175. $4(2.5v - 0.6) = 7.6$

176. $0.2(30n + 50) = 28$

177. $0.5(16m + 34) = -15$

178. $-(w - 6) = 24$

179. $-(t - 8) = 17$

180. $9(3a + 5) + 9 = 54$

181. $8(6b - 7) + 23 = 63$

182. $10 + 3(z + 4) = 19$

183. $13 + 2(m - 4) = 17$

184. $7 + 5(4 - q) = 12$

185. $-9 + 6(5 - k) = 12$

186. $15 - (3r + 8) = 28$

187. $18 - (9r + 7) = -16$

188. $11 - 4(y - 8) = 43$

189. $18 - 2(y - 3) = 32$

190. $9(p - 1) = 6(2p - 1)$

191. $3(4n - 1) - 2 = 8n + 3$

192. $9(2m - 3) - 8 = 4m + 7$

193. $5(x - 4) - 4x = 14$

194. $8(x - 4) - 7x = 14$

195. $5 + 6(3s - 5) = -3 + 2(8s - 1)$

196. $-12 + 8(x - 5) = -4 + 3(5x - 2)$

197. $4(x - 1) - 8 = 6(3x - 2) - 7$

198. $7(2x - 5) = 8(4x - 1) - 9$

Everyday Math

199. **Making a fence** Jovani has a fence around the rectangular garden in his backyard. The perimeter of the fence is 150 feet. The length is 15 feet more than the width. Find the width, w, by solving the equation $150 = 2(w + 15) + 2w$.

200. **Concert tickets** At a school concert, the total value of tickets sold was \$1,506. Student tickets sold for \$6 and adult tickets sold for \$9. The number of adult tickets sold was 5 less than 3 times the number of student tickets. Find the number of student tickets sold, s, by solving the equation $6s + 9(3s - 5) = 1506$.

201. **Coins** Rhonda has \$1.90 in nickels and dimes. The number of dimes is one less than twice the number of nickels. Find the number of nickels, n, by solving the equation $0.05n + 0.10(2n - 1) = 1.90$.

202. **Fencing** Micah has 74 feet of fencing to make a rectangular dog pen in his yard. He wants the length to be 25 feet more than the width. Find the length, L, by solving the equation $2L + 2(L - 25) = 74$.

Writing Exercises

203. When solving an equation with variables on both sides, why is it usually better to choose the side with the larger coefficient as the variable side?

204. Solve the equation $10x + 14 = -2x + 38$, explaining all the steps of your solution.

205. What is the first step you take when solving the equation $3 - 7(y - 4) = 38$? Explain why this is your first step.

206. Solve the equation $\frac{1}{4}(8x + 20) = 3x - 4$ explaining all the steps of your solution as in the examples in this section.